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(54) FORCE TRANSMISSION MECHANISMS AND RELATED DEVICES AND METHODS

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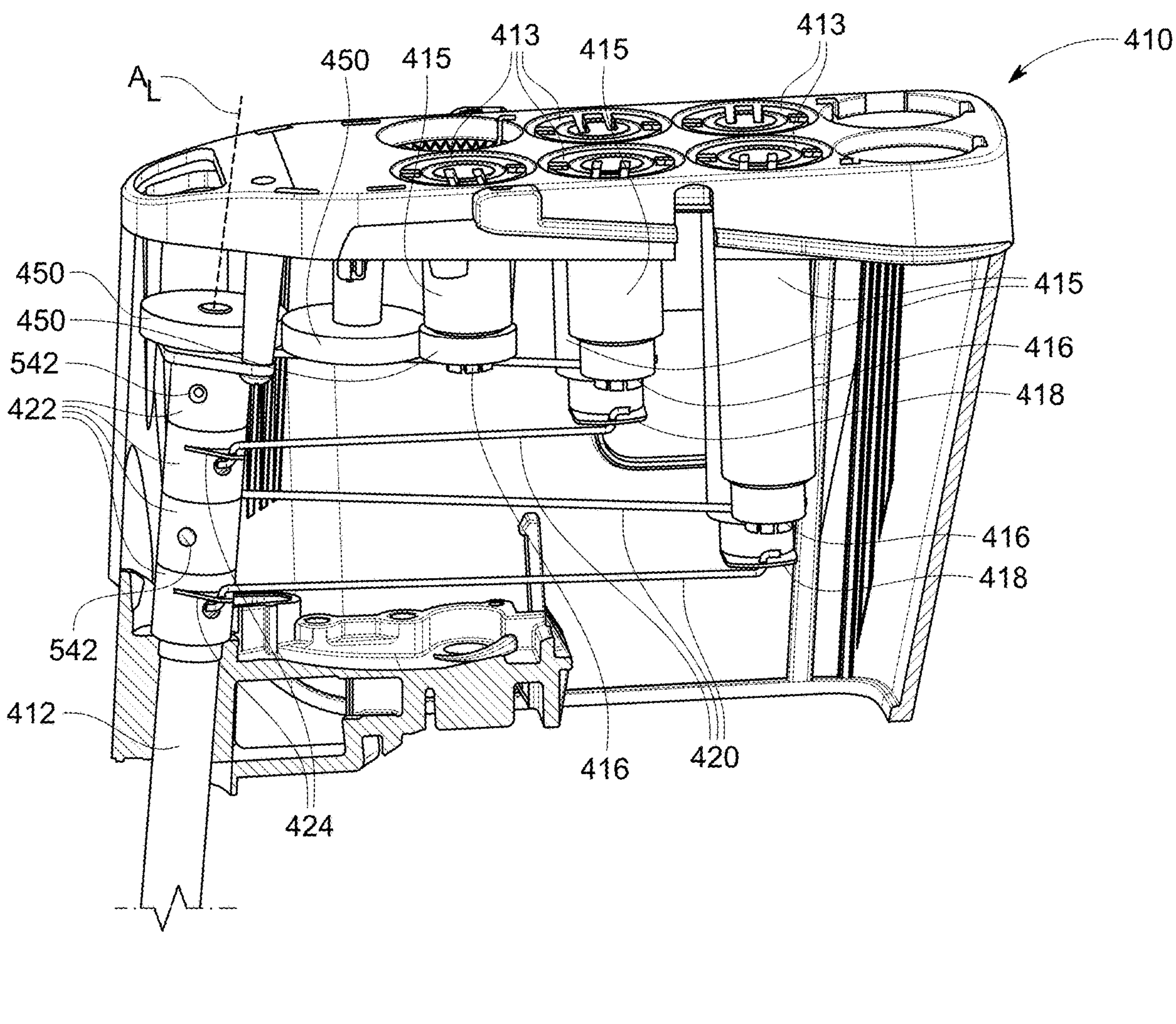
(57) ABSTRACT

A steerable instrument includes a shaft comprising an articulable segment; a force transmission mechanism at the proximal end portion of the shaft, the force transmission mechanism comprising one or more rotatable drive components having axes of rotation spaced from the longitudinal axis of the shaft, the one or more rotatable drive components configured to be rotatably driven by respective drive input torques; one or more push-pull actuation elements configured to translate to transmit compressive forces to the articulable segment; and a rotatable coupler mechanism coupling one of the one or more rotatable drive components to the one or more push-pull actuation elements, the rotatable coupler mechanism configured to rotate about the longitudinal axis of the shaft in response to rotation of the one of the one or more rotatable drive components to cause translation of the one or more push-pull actuation elements.



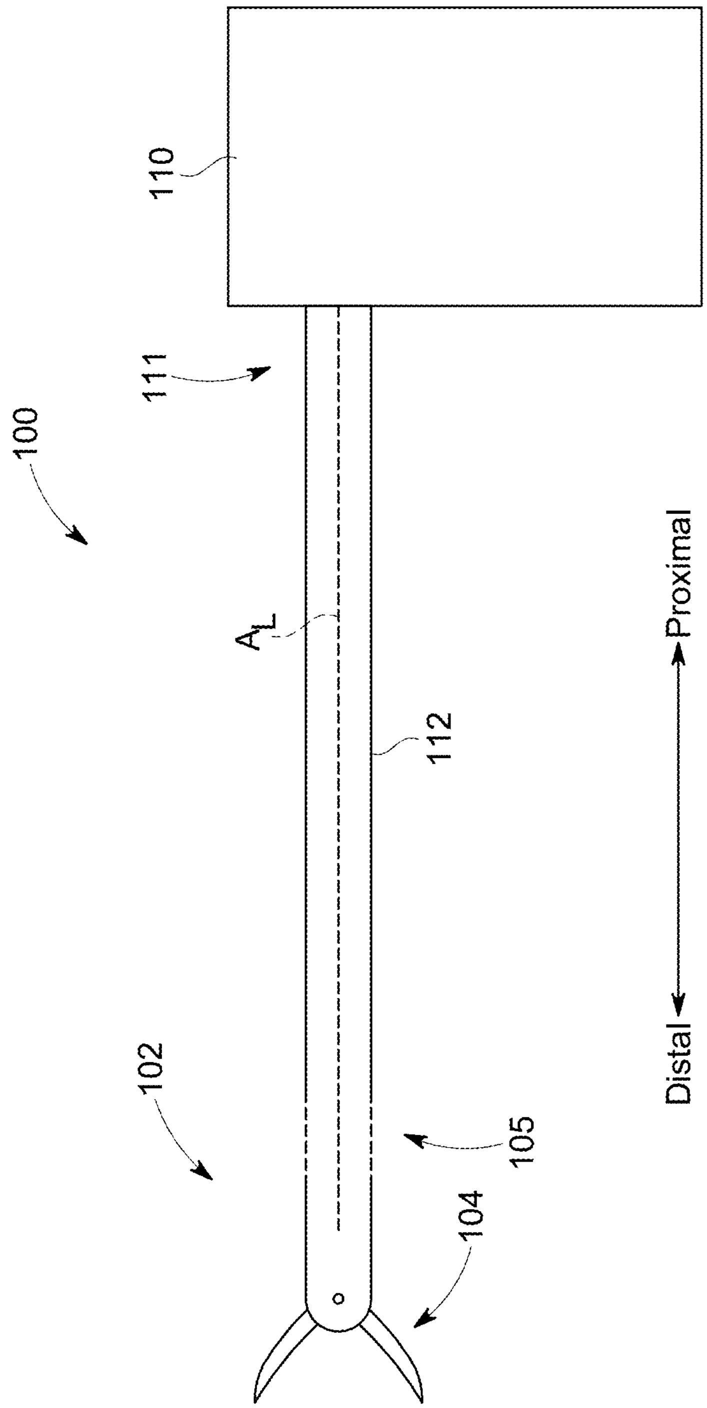


FIG. 1

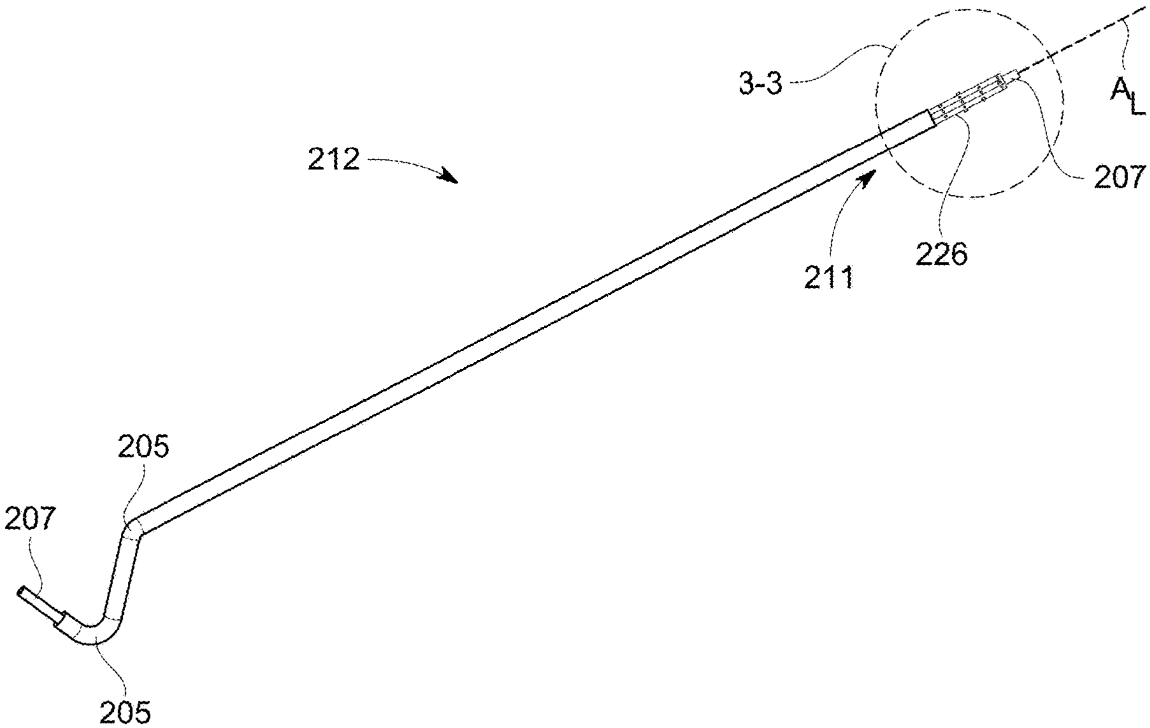

3-3
212
207
226
211
205
207
205
A_L

FIG. 2

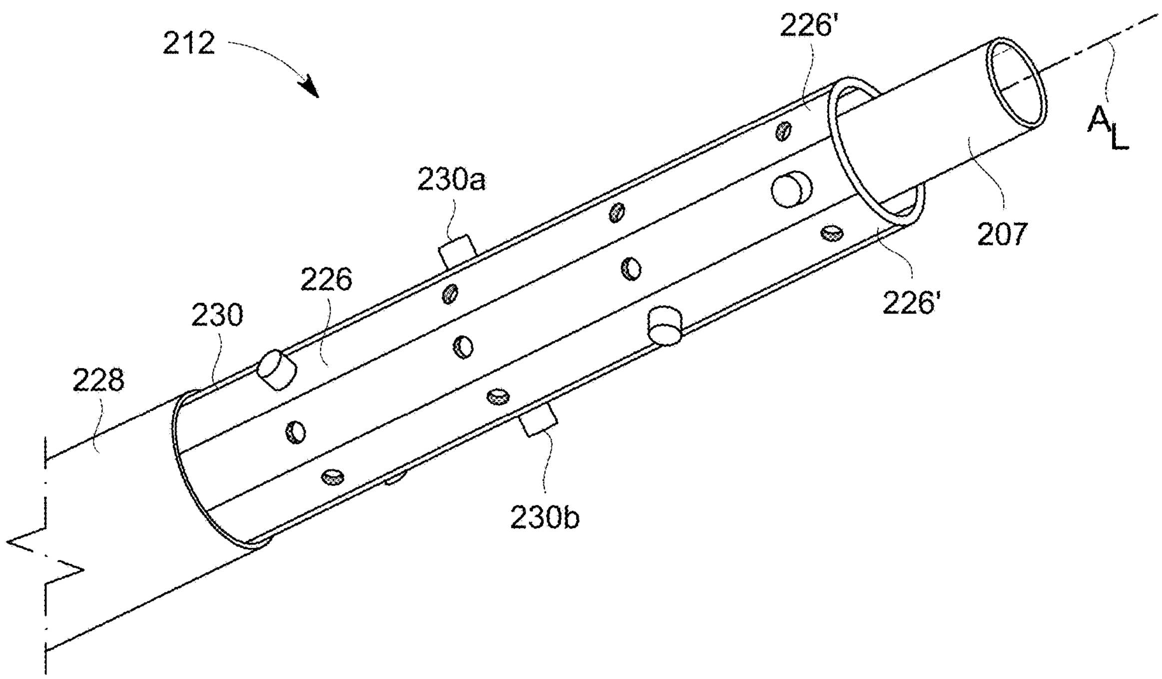

212
226'
230a
207
226
226'
230
228
230b
A_L

FIG. 3

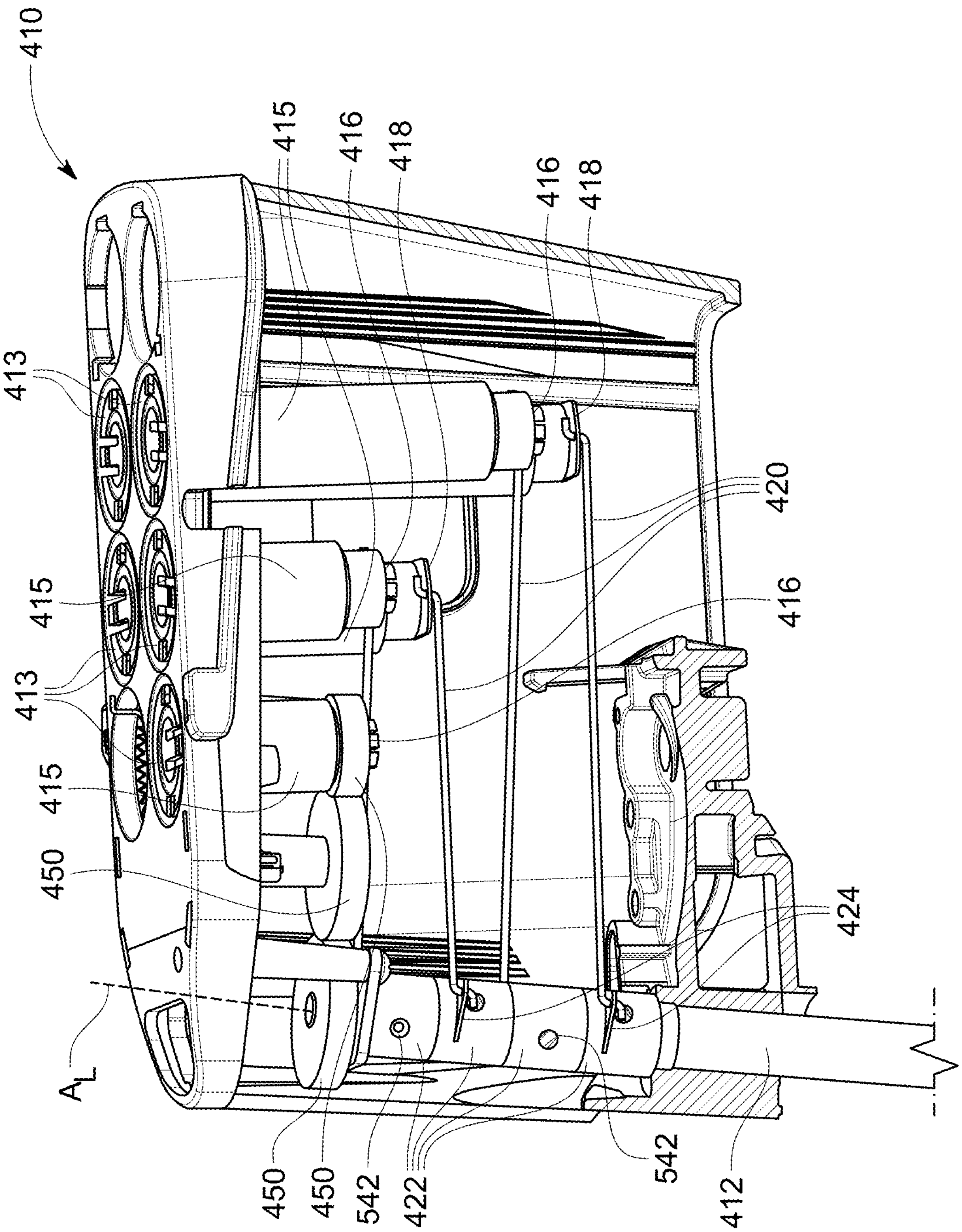

410
413
413
415
415
415
416
418
416
418
420
416
450
A_L
450
450
542
422
542
412
424

FIG. 4

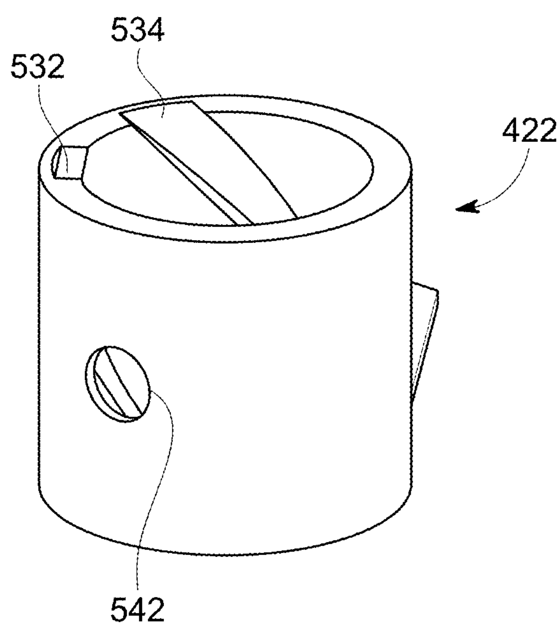


FIG. 5A

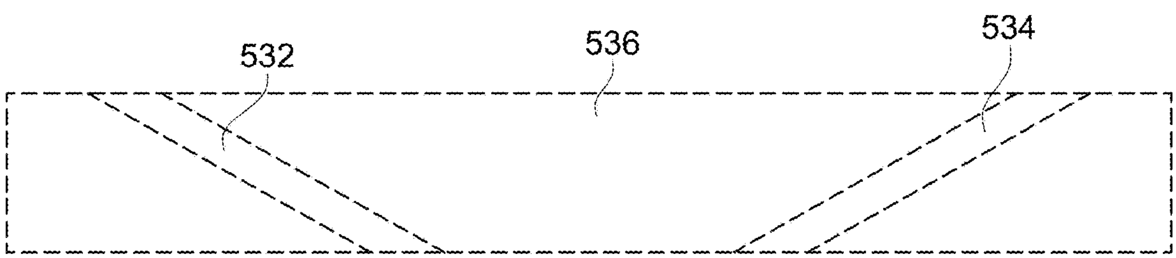


FIG. 5B

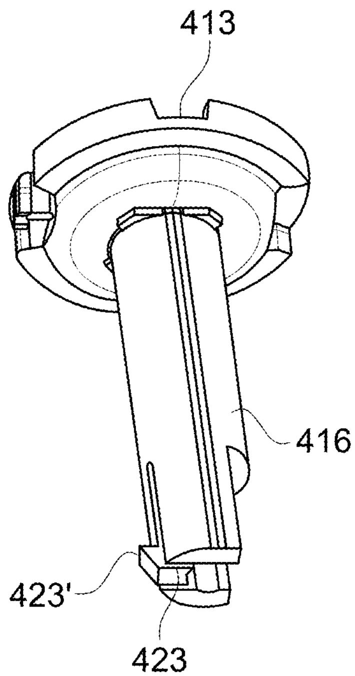


FIG. 6

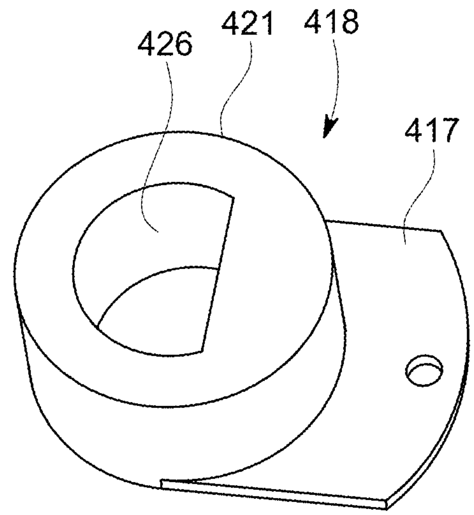


FIG. 7

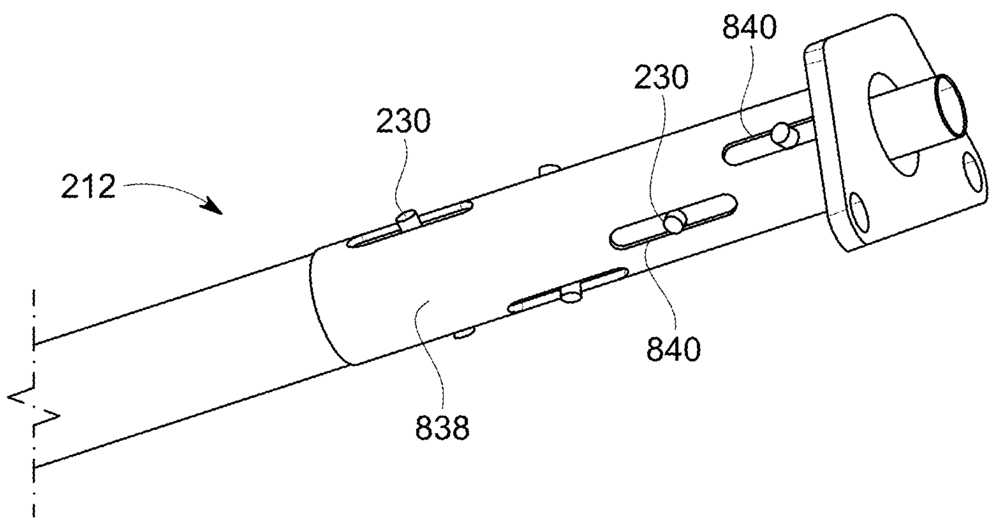


FIG. 8

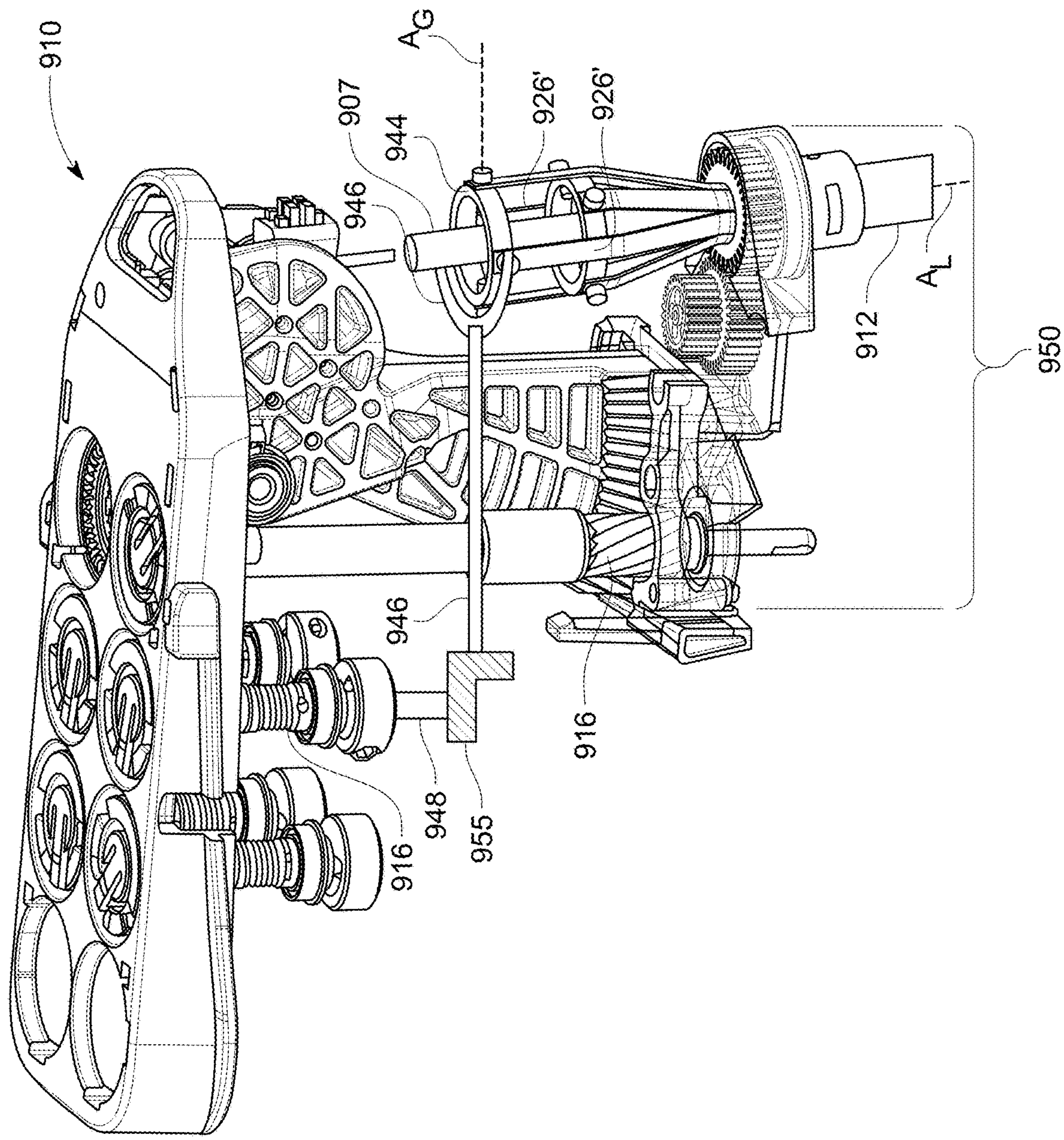


FIG. 9

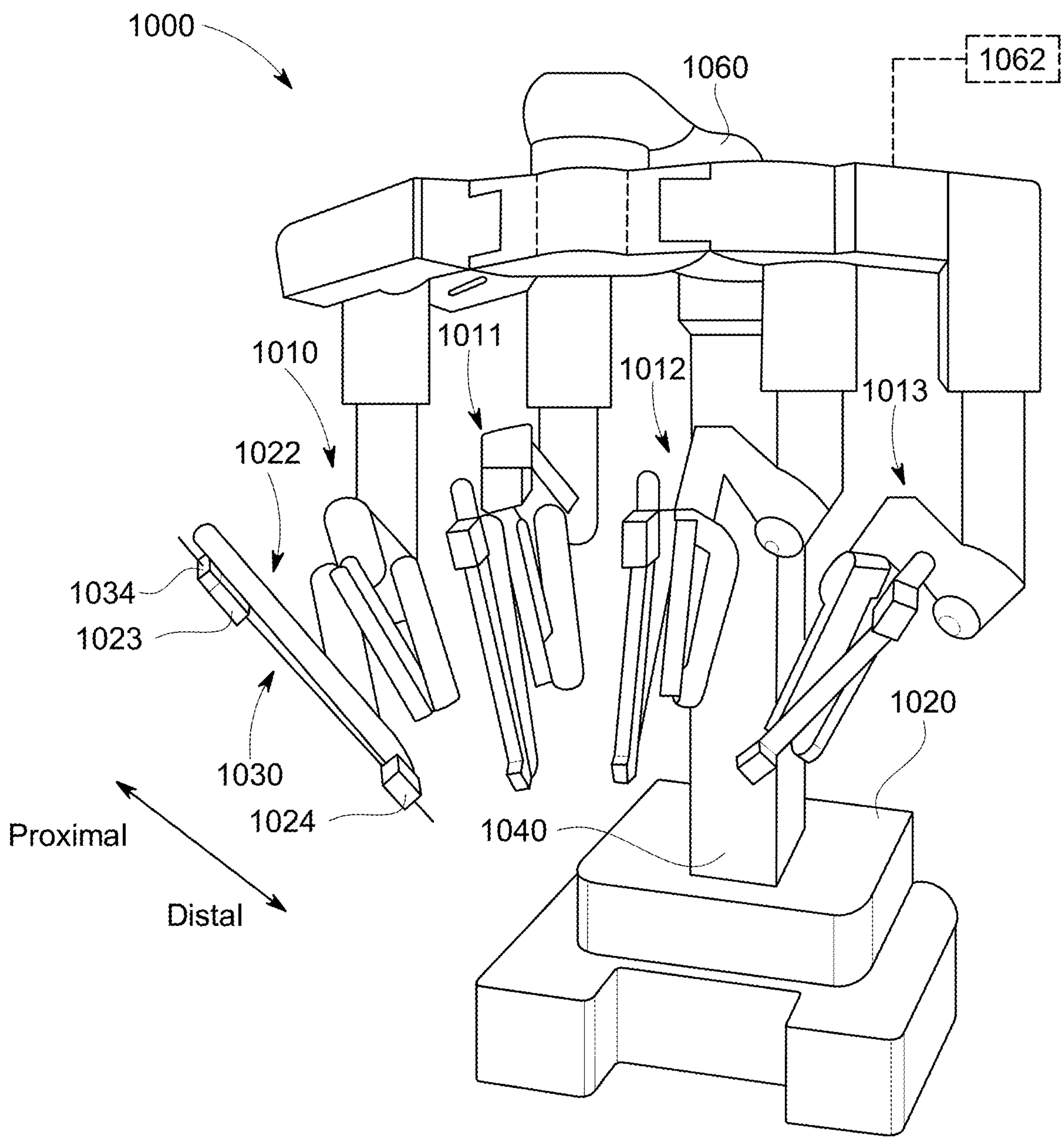

1000
1060
1062
1010
1011
1012
1013
1022
1034
1023
1030
1024
1020
1040
Proximal
Distal

FIG. 10

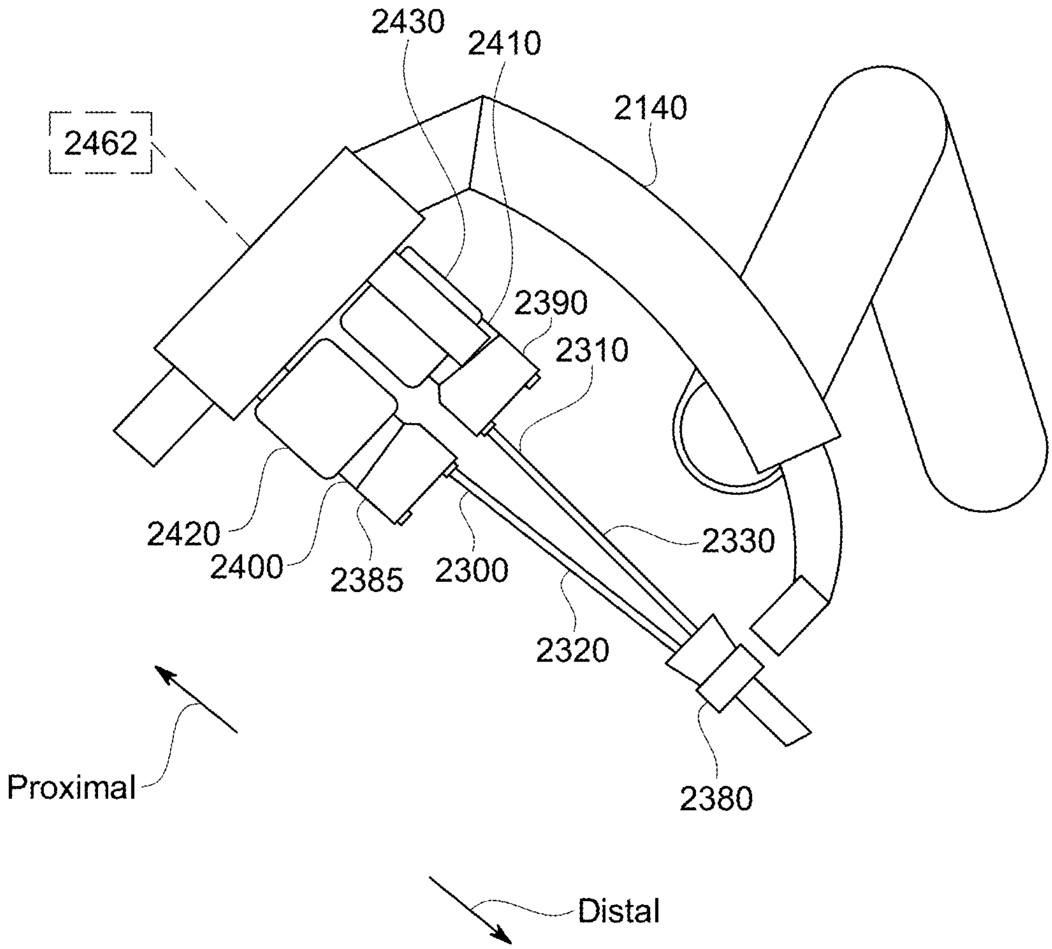

2430
2410
2140
2462
2390
2310
2420
2400
2385
2300
2330
2320
2380
Proximal
Distal

FIG. 11

FORCE TRANSMISSION MECHANISMS AND RELATED DEVICES AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Application No. 63/559,300, filed Feb. 29, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] Aspects of the present disclosure relate to force transmission mechanisms and related devices and methods. For example, aspects of the present disclosure relate to force transmission mechanisms that convert rotational input forces to translational forces that articulate and/or actuate components of an instrument, such as a medical or other remotely actuatable instrument.

INTRODUCTION

[0003] Various tools such as medical (including surgical) or industrial instruments often include shafts having one or more moveable structures along a length of the shaft or coupled to an end of the shaft. Such moveable structures can include joints or end effectors that impart one or more degrees of freedom of movement to such instruments. Such moveable structures are often actuated and controlled via translating actuation elements extending along a length of the shaft and coupled to the moveable structure at one end and to a force transmission mechanism at an opposite end. Such actuation elements may include, for example, cables or other tension members that operate in a pull-pull manner to transmit tensile (pulling) force acting on the member so as to actuate the moveable structure, or rods or other compression members that operate in a push-pull manner to transmit tensile (pulling) or compressive (pushing) force acting on the member so as to actuate the moveable structure.

[0004] Force transmission mechanisms are generally situated at a proximal end portion of the shaft of an instrument, with the end effector of the instrument situated at a distal end portion. Force transmission mechanisms can include a variety of driven force transmission components, such as capstans, gears, linkages, levers, and the like coupled to drive inputs that can be manually operated or configured to interface with a computer-assisted, teleoperated manipulator system.

[0005] In various applications, force transmission mechanisms include rotary input drive members (sometimes referred to as input drive disks). Rotation of rotary input drive members is then converted to translation of the actuation elements through various components and structures configured to transform rotational drive forces to translational forces to translate the actuation elements, thereby transmitting the desired force to articulate a moveable component of instrument.

[0006] Due to various architectural factors and packaging constraints, some instrument force transmission mechanisms have limited space. Moreover, some instrument force transmission mechanisms may have input drive members that are laterally offset from the instrument shaft and the actuation elements that are to be driven by the input drive members. In addition, the amount of force and force conversion needed in some of the limited space, relatively high friction environments found in some instruments pose challenges for robust and durable force transmission mechanisms.

[0007] There exists a need for force transmission mechanisms that are compact, mechanically robust, and able to be adapted to existing instrument shaft and manipulator designs. Additionally, there exists a need for such systems that possess desirable characteristics such as manufacturability and reliability in operation.

SUMMARY

[0008] Embodiments of the present disclosure may solve one or more of the above-mentioned problems and/or may demonstrate one or more of the above-mentioned desirable features. Other features and/or advantages may become apparent from the description that follows.

[0009] In accordance with at least one aspect of the present disclosure, A steerable instrument includes a shaft extending along a longitudinal axis from a proximal end portion to a distal end portion, the shaft comprising an articulable segment; a force transmission mechanism at the proximal end portion of the shaft, the force transmission mechanism comprising one or more rotatable drive components having axes of rotation spaced from the longitudinal axis of the shaft, the one or more rotatable drive components configured to be rotatably driven by respective drive input torques; one or more push-pull actuation elements coupled to the articulable segment and configured to translate to transmit compressive forces to the articulable segment to articulate the articulable segment; and a rotatable coupler mechanism coupling one of the one or more rotatable drive components to the one or more push-pull actuation elements, the rotatable coupler mechanism configured to rotate about the longitudinal axis of the shaft in response to rotation of the one of the one or more rotatable drive components to cause translation of the one or more push-pull actuation elements.

[0010] In another aspect of the present disclosure, a steerable instrument includes a shaft extending along a longitudinal axis from a proximal end portion to a distal end portion, the shaft comprising an articulable segment; a force transmission mechanism at the proximal end portion of the shaft, the force transmission mechanism comprising a rotatable drive component having an axis of rotation spaced from the longitudinal axis of the shaft, the rotatable drive component configured to be rotatably driven by a drive input torque; a pair of push-pull actuation elements coupled to the articulable segment and configured to translate so as to transmit compressive forces to the articulable segment to articulate the articulable segment; and a rotatable coupler mechanism coupling the rotatable drive component to the pair of push-pull actuation elements, the rotatable couple mechanism configured to rotate in response to rotation of the rotatable drive component to cause translation of the pair of push-pull elements in opposite directions.

[0011] In yet another aspect of the present disclosure, a steerable instrument includes a shaft extending along a longitudinal axis from a proximal end portion to a distal end portion, the shaft comprising an articulable segment; a first actuation element operably coupled to the articulable segment of the shaft and extending along the shaft from the articulable segment to the proximal end portion of the shaft; a second actuation element operably coupled to the articulable segment of the shaft and extending along the shaft from the articulable segment to the proximal end portion of the shaft; and a rotatable coupler mechanism rotatable about \the longitudinal axis of the shaft and comprising cam surfaces respectively engageable with the first and second actuation elements so as to drive the first and second actuation elements in translation in response to rotation of the rotatable coupler mechanism.

[0012] Yet another aspect of the present disclosure contemplates a method of articulating an articulable segment of an instrument shaft that includes driving rotation of a rotatable coupler mechanism via a rotatable drive component having an axis of rotation offset from an axis of rotation of the rotatable coupler mechanism, wherein the rotatable coupler mechanism is engaged with a push-pull actuation element coupled to the articulable segment; and in response to rotation of the rotatable coupler mechanism, driving translation of the push-pull actuation element so as to transmit a compressive force by the push-pull actuation element to cause articulation of the articulable segment.

[0013] Additional objects, features, and/or advantages will be set forth in part in the description which follows, and in part will be obvious from the description, or may be learned by practice of the present disclosure and/or claims. At least some of these objects and advantages may be realized and attained by the elements and combinations particularly pointed out in the appended claims.

[0014] It is to be understood that both the foregoing general description and the following detailed description are for example and explanatory only and are not restrictive of the claims; rather the claims should be entitled to their full breadth of scope, including equivalents.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings are included to provide a further understanding of the present disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate one or more embodiments of the present teachings and together with the description explain certain principles and operation. In the drawings,

[0016] FIG. **1** is a schematic, side view of an embodiment of an instrument comprising a force transmission mechanism.

[0017] FIG. **2** is a perspective view of an embodiment of a shaft of an instrument.

[0018] FIG. **3** is an enlarged view of a proximal end portion of the shaft of FIG. **2**.

[0019] FIG. **4** is an interior view of a force transmission mechanism according to an embodiment.

[0020] FIG. **5**A is a perspective view of an embodiment of a rotatable sleeve of the force transmission mechanism of FIG. **4**.

[0021] FIG. **5**B is a projection view of an interior surface of the rotatable sleeve of FIG. **5**A.

[0022] FIG. **6** is a perspective view of a drive input of the force transmission mechanism of FIG. **4**.

[0023] FIG. **7** is a perspective view of an embodiment of a crank arm of the force transmission mechanism of FIG. **4**.

[0024] FIG. **8** is a perspective view of another embodiment of an instrument shaft.

[0025] FIG. **9** is a perspective, interior view of a force transmission mechanism according to another embodiment.

[0026] FIG. **10** is a perspective view of an embodiment of a manipulator system.

[0027] FIG. **11** is a partial schematic view of another embodiment of a manipulator system.

DETAILED DESCRIPTION

[0028] Embodiments of the present disclosure relate to force transmission mechanisms that are configured to transmit drive inputs, received from computer-assisted (e.g., teleoperated) manipulators or manually-operated input devices, to forces to actuate actuation elements operably coupled to control movement of moveable components of an instrument, such as joints along an instrument shaft or an end effector. Force transmission systems according to various embodiments of the present disclosure enable drive input forces to occur at a location that is laterally offset from the longitudinal axis of the instrument shaft. Further, various embodiments of force transmission mechanisms can allow for transmission of force to multiple actuation elements in a relative limited and compact space. In addition, some embodiments of force transmission mechanisms according to the present disclosure include drive components that are configured to transmit drive forces from multiple drive inputs, each of which is configured to rotate about a respective axis, to actuation elements of a shaft arranged about a longitudinal axis of the shaft offset from the location of the axes of the drive inputs.

[0029] Embodiments of transmission systems according to the present disclosure may have particular relevance to shafts including non-cable type actuation elements, such as push-pull actuation elements. For example, some types of articulable shafts include concentric longitudinal tubular members that may be formed out of one or more layers of sheet metal. The articulable shafts may further include push-pull actuation elements formed or cut out of the one or more layers of sheet metal. Such push-pull actuation elements, unlike tension member (pull-pull type) actuation elements (such as cables or the like), generally cannot be routed around pulleys, capstans, etc. to easily change direction and route along a particular pathway dictated by the overall layout of the force transmission mechanism and instrument shaft. However, push-pull actuation elements may have certain advantages as compared to tension member (pull-pull type) actuation elements, such as but not limited to greater resistance to buckling and therefore the ability to transmit pushing, as well as pulling, force to a coupled component, and lower cost of manufacturing and/or assembly in an instrument. Various embodiments of transmission systems as disclosed herein may facilitate adaptation of shaft designs, including those incorporating layered sheets of push-pull type actuation elements to manipulator system designs typically used in conjunction with tension member (pull-pull) actuated instruments.

[0030] Force transmission systems of various embodiments may include mechanical drive component arrangements that transmit drive forces in the form of rotational forces about the axes of the drive inputs to a common axis of the instrument shaft about which the actuation elements are arranged. In some arrangements of instruments and associated manipulators, the axes of the drive inputs can be laterally offset from the common axis of the instrument shaft. In embodiments of the disclosure in which a single degree of freedom of movement of the shaft is actuated via two actuation elements moving in opposite directions (such as the use of a pair of opposing cables to effect pitch or yaw of a distal portion of the shaft), force transmission mechanisms of the present disclosure can provide simultaneous, opposing movement of each actuation element based on a single input, e.g., a rotational drive input. Embodiments of the disclosure provide such configurations under the space constraints imposed by some manipulator designs.

[0031] Referring now to FIG. **1**, a schematic, side view of an elongate instrument **100** according to some embodiments is shown. Instrument **100** can be or include an instrument used to perform surgical, diagnostic, therapeutic, and other medical or non-medical procedures. The instrument **100** includes an end effector **104**, a shaft **112** defining a longitudinal axis A_L, and a force transmission mechanism **110**. The end effector **104** is located at a distal end portion **102** of the shaft **112**. The end effector **104** can be configured to carry out a medical or non-medical (such as industrial) procedure. For example, the end effector **104** can include one or more tools such as gripping tools, staplers, shears, ligation clip appliers, electrosurgical tools, or other types of tools. While the illustration of FIG. **1** depicts an end effector having openable/closeable jaw members, such a configuration is exemplary and non-limiting and those of ordinary skill in the art would appreciate the instrument **100** can have any of a variety of end effectors without departing from the scope of the present disclosure

[0032] In the embodiment of FIG. **1**, the force transmission mechanism **110** is coupled to a proximal end portion **111** of the shaft **112**. In other embodiments, the force transmission mechanism **110** may be coupled at a mid-portion of the shaft **112**, such as distal to a proximal end of the shaft **112**. The force transmission mechanism **110** can be configured to be operably coupled with (e.g., removably coupled with) a computer-assisted (e.g., teleoperated) manipulator system, such as the manipulator systems described in further detail below in connection with FIGS. **10** and **11** or similar manipulator systems with which those having ordinary skill in the art are familiar. In other embodiments, in addition to or in lieu of being configured to interface and be driven by a computer-assisted manipulator system, the force transmission mechanism **110** can be manually controlled with manually operated (e.g., handheld) actuators (not shown).

[0033] In the embodiment shown in FIG. **1**, the instrument **100** includes an articulable segment **105** arranged along the shaft **112** between the end effector **104** and the force transmission mechanism **110**. As shown in FIG. **1**, the articulable segment **105** can be positioned toward the distal end portion **102** of the shaft **112**. However, the disclosure is not so limited and the articulable segment **105** can be positioned at any location along the shaft **112** without limitation. In addition, the instrument **100** can include more than one articulable segment **105**, such as two, three, or more articulable segments located at multiple locations along the length of the shaft **112**. The articulable segment **105** can be controlled and actuated via actuation elements (not shown in FIG. **1**; discussed above and in connection with the various embodiments of FIGS. **2-9**) operably coupled to a manipulator by a force transmission mechanism. The articulable segment **105** can include one or more joints configured as flexible portions (e.g., sections of the shaft in which the material of the shaft is deflectable) of the shaft **112**, or as structural joints (e.g., hinged or otherwise pivotable portions) of the shaft **112**. The one or more articulable segments **105** can be articulated by actuation and control of the actuation elements to provide pitch and/or yaw motion to the end effector **104**.

[0034] In some embodiments as discussed herein, the shaft **112**, the one or more articulable segments **105**, and actuation elements housed in the shaft **112** (not shown in FIG. **1**; discussed in connection with the various embodiments of FIGS. **2-9**) may have a configuration like that disclosed at least in, for example, U.S. Pat. No. 8,740,884 (issued Jun. 3, 2014) titled "INSTRUMENT FOR ENDOSCOPIC APPLICATIONS AND THE LIKE," (the '884 patent); U.S. Pat. No. 8,986,317 (issued Mar. 24, 2015) titled "INSTRUMENT AND METHOD FOR MAKING THE SAME"; U.S. Patent App. Pub. No. 2012/0245414 (filed Mar. 23, 2011) titled "HANDLE FOR CONTROLLING INSTRUMENTS, ENDOSCOPIC INSTRUMENT COMPRISING SUCH A HANDLE, AND AN ASSEMBLY," (the '414 publication), and U.S. Patent App. Pub. No. 2018/0008805 (filed Jun. 5, 2017) titled "METHOD FOR MANUFACTURING A STEERABLE INSTRUMENT AND SUCH A STEERABLE INSTRUMENT," (the '805 publication), the entire contents of each of which are incorporated herein by reference. For example, the longitudinal elements **4** disclosed in the '805 application can correspond to actuation elements as discussed herein. Embodiments of the shafts disclosed in the patent and publications identified above can include push-pull type actuation elements formed of one or more layers of metal or other material, such as a laminated sheet metal actuation element, as discussed above. Moreover, other embodiments of shafts, such as those including tension member (pull-pull type) actuation elements and not related to the patent and publications identified above, are within the scope of the present disclosure.

[0035] Referring now to FIG. **2**, an embodiment of a shaft **212** that can be used as shaft **112** is shown in isolation to better illustrate various aspects of the shaft **212**. The shaft **212** extends generally along a longitudinal axis A_L and includes one or more articulable segments **205**. As shown in FIG. **2**, the articulable segments **205** can be located at a distal end portion **202** of the shaft **212**. FIG. **2** is provided as exemplary in nature and in other embodiments, one or more articulable segments **205** can be positioned further proximally along the shaft, and any desired number and position of articulable segments is considered within the scope of the disclosure.

[0036] The distal end portion **202** of the shaft **212** can include an end effector (not shown) such as, but not limited to, tools such as grippers, staplers, shears, ligation clip appliers, electrosurgical tools, imaging tools (e.g., endoscopes), or other types of tools as described above with respect to the embodiment of FIG. **1**. In some embodiments, the articulable segments **205** can provide pitch and/or yaw degrees of freedom, and the movement of the articulable segments **205** can be controlled to provide pitch and/or yaw motion to the end effector relative to the proximal portion of the shaft **212**. The articulable segment **205** can be actuated to impart a desired geometry to the shaft **212** for accessing worksites, such as sites of interest in minimally invasive medical procedures. The shaft **212** can comprise multiple tubular elements or layers, including an inner layer **207**, an intermediate layer **226**, and an outer layer **228**. The layers may have tubular configurations, such as hollow cylindrical configurations. The intermediate layer **226** is located radially between the inner layer **207** and the outer layer **228**. The intermediate layer **226** includes push-pull actuation elements **226'** for controlling the instrument, such as those described above with respect to FIG. **1**. In some embodiments, the intermediate layer itself may be further subdivided into multiple thinner layers having a laminate structure, which can facilitate the ability to bend with less bending stress and force, with each of the multiple thinner layers having one or more actuation elements **226'**. The inner layer **207** and outer layer **228** provide structure and form to the shaft **212** and can comprise flexible segments throughout and/or only at the locations of the articulable segments **205**, with rigid segments in other areas of the shaft **212**. In some embodiments, reliefs in the layers can be used to affect rigidity.

[0037] An end effector (not shown in FIG. **2**; e.g., end effector **104** shown in FIG. **1**) can be coupled to the inner layer **207**. The inner layer **207** is coupled to a force transmission mechanism (not shown), such as force transmission mechanism **110** of FIG. **1**), and can be actuated to roll about the longitudinal axis A_L by the force transmission mechanism, and in turn impart a roll degree of freedom movement to the end effector about the longitudinal axis A_L. Inputs from a manipulator to which the force transmission mechanism is coupled can be provided to move one or more drive components of the force transmission mechanism to actuate the roll degree of freedom. In various embodiments disclosed herein, the inner layer **207** is rotationally decoupled from the intermediate layer **226** and the outer layer **228** such that rotation of the inner layer **207** about the longitudinal axis A_L does not cause a corresponding rotation of either the intermediate layer **226** (or the actuation elements **226'** thereof) or of the outer layer **228**. Thus, the roll orientation of the inner layer **207** (and thus the orientation of the end effector) does not impact the overall configuration of the articulable segments **205** of the shaft **212**. The inner layer **207** can have one or more central lumens (not shown) through which one or more actuation elements (not shown and other than actuation elements **226'**), such as push-pull (e.g., compression members) and/or pull-pull (e.g., tension members) actuation elements, can be routed for actuating the end effector, as those having ordinary skill in the art are familiar with.

[0038] Control of the articulable segments **205** can be accomplished by application of translational forces to actuation elements within the shaft **212**. As discussed above, the shaft **212** can include push-pull type actuation elements. In the embodiment of FIG. **2**, actuation elements **226'** are exposed at the proximal end portion **211** of the shaft **212**. FIG. **3** shows an enlarged view of the proximal end portion **211** of the shaft **212** and illustrates actuation elements **226'** arranged around the central longitudinal axis A_L of the shaft **212**. The actuation elements **226'** can comprise, for example, actuation elements formed of a single layer or multiple laminated layers of sheet metal, as discussed in the '884 patent for example. Other types of actuation elements, such as pull-pull tension actuation elements (e.g., cables), push-pull compression member actuation elements (e.g., rods), and other elements can also be actuated using the force transmission mechanisms in accordance with various embodiments.

[0039] In various implementations, an articulable segment of a shaft can be articulated by actuation of a push-pull actuation element. In the embodiment of FIG. **2**, each degree of freedom of the shaft **212** is associated with a pair of push-pull actuation elements **226'** located diametrically opposite one another relative to the axis A_L. Using such a pair of push-pull actuation elements **226'** can allow for one of the actuation elements **226'** to transmit a push force while the other transmits a pull force, thereby reducing the load on the main shaft. For example, a shaft including two articulable segments **205** (FIG. **2**) each of which is separately actuatable in two degrees of freedom relative to the axis A_L (such as movement in pitch and yaw, respectively) may thus include eight actuation elements **226'** positioned around the shaft, as described further with reference to of the embodiment of FIG. **3**. A pair of actuation elements **226'** generates movement in a single degree of freedom, such as one of pitch or yaw of one of the articulable segments **205**, in response to opposing forces applied to the pair of actuation elements **226'**. In other words, for each pair of actuation element **226** assigned to each degree of freedom, application of a distally-directed force (push force) to one actuation element **226** of the pair and a proximally-directed force (pull force) to the other actuation element **226** of the pair generates articulation of the articulable segment of the shaft in that degree of freedom. In some embodiments, movement of each degree of freedom is controlled by each pair of diametrically oppositely located actuation elements, but embodiments the present disclosure are not so limited. For example, the actuation elements could be spaced around the shaft **212** such that each degree of freedom is controlled by actuation elements spaced other than 180 degrees apart with respect to the shaft **212**. Moreover, as noted above, any number of actuation elements can be used from 1 to more than 2 for actuation of an articulable segment and may be selected based on factors such as, but not limited to, overall load, space considerations, and complexity.

[0040] As discussed above, each actuation element **226'** is part of an intermediate layer **226** positioned between the outer layer **228** and the inner layer **207**. Each actuation element **226'** is exposed from (extends beyond) the outer layer **228** at the proximal end portion **211** of the shaft **212** and extends distally to the articulable segment **205** with which the particular actuation element **226'** is associated. Each actuation element **226'** includes one or more features configured to interface with the force transmission mechanism to receive application of push or pull force to articulate the shaft **212** at the articulable segments **205**.

[0041] For example, in FIG. **3**, each of the actuation elements **226'** includes an engagement feature **230**. As shown in FIG. **3**, the engagement features **230** of different opposing pairs of actuation elements **226'** are staggered along the longitudinal axis A_L of the shaft **212**. That is, with reference to a pair of engagement features **230*a***, **230*b*** of an opposing pair of actuation elements **226'**, the engagement features **230*a***, **230*b*** are aligned along the axial direction (i.e., positioned diametrically opposite each other), but are spaced along the axial direction (not aligned) from engagement features **230** of the other opposing pairs of actuation elements **226'**, and likewise for the engagement features **230** for each pair of opposing actuation elements **226'**. Each engagement feature **230** is configured to engage with a component of the force transmission mechanism to receive actuation force from the transmission system, as discussed further below. As shown in FIG. **3**, the engagement features **230** are in the form of protrusions, such as protruding pins configured to cooperate with a cam surface of the force transmission mechanism as is described further below.

[0042] Referring now to FIG. **4**, an embodiment of a force transmission mechanism **410** is shown. The force transmission mechanism **410** includes a chassis **414** to which the various components are coupled, and which is configured to couple to an instrument manipulator system (such as but not limited to manipulator systems as disclosed in connection with FIGS. **10** and **11** herein). In other embodiments (not illustrated), a force transmission mechanism can be configured with the internal drive components described for force transmission mechanism **410** but be modified to be driven via manual input mechanisms for use with manually operated instruments as those of ordinary skill in the art are familiar with. The force transmission mechanism **410** includes rotatable drive components in the form of drive input shafts **416** extending through shaft supports **415** of the chassis **414** and coupled to rotary drive disks at an exterior of the chassis **414**. In an installed state of the force transmission mechanism **410** on a manipulator system, the input rotary drive discs, and thereby the drive input shafts **416**, are configured to be operably coupled to drive outputs associated with the manipulator system, such as output drive discs or shafts. Rotation of the drive outputs of the manipulator system drives rotation of the input rotary drive disks and drive input shafts **416** of the force transmission mechanism **410**. Also shown in FIG. **4** is a series of gears **450** (shown without teeth) that transmit torque from drive input shaft **416** to provide a roll degree of freedom to the inner layer **207** (FIG. **2**) based on inputs from a manipulator system or manually, as discussed above.

[0043] As discussed above, the instrument shaft **412** is positioned along a central longitudinal axis A_L offset from the rotational axes of the respective drive input shafts **416**. To impart motion to the instrument shaft **412** and end effector coupled at the distal end portion thereof (not shown in FIG. **4**), motion of the drive input shafts **416** must be transferred from the drive input shafts **416** to actuation elements of the shaft **412**. In the embodiment of FIG. **4**, a mechanical linkage system is used to transfer the rotational motion of the drive input shafts **416** from rotational axes of the drive input shafts **416** to the actuation elements of the shaft **412**, the longitudinal axis A_L of which is laterally spaced from the individual rotational axes of the drive input shafts **416**.

[0044] The mechanical linkages of the embodiment of FIG. **4** comprise a respective crank arm **418** (two of which are labeled in FIG. **4**) to which each of the drive input shafts **416** is coupled. An embodiment of a crank arm **418** is shown in isolation in FIG. **7** and is described further below. Crank arm **418** is coupled to a first end of a connecting rod **420**, with a second end of the connecting rod **420** being operably coupled to the instrument shaft **412** via a rotatable sleeve **422**, shown in isolation in FIG. **5** and also described further below. The crank arm **418** is configured to be rotationally fixed with the drive input shaft **416** and to be held axially in position on the shaft **416**.

[0045] The interface between the crank arm and the shaft can have various arrangements, such as, but not limited to splined interfaces, keyed interfaces, and any other suitable hub-shaft type interface so as to provide a fixed rotational relationship between the shaft **416** and the crank arm **418**. Further, in some embodiments, the crank arms **418** could be made integral with the drive input shafts **416**, or provided in any manner familiar to those having skill in the art.

[0046] The mechanical linkage components of the force transmission mechanism **410** further include one or more rotatable coupler mechanism that are coupled to a pair of actuation elements **226'** (FIGS. **2** and **3**) such that rotation of a rotatable coupler mechanism causes translation of a pair of actuation elements **226'** in opposite directions to each other. For example, referring again to FIG. **4**, a plurality of rotatable coupler mechanisms are provided in the form of rotatable sleeves **422** arranged in series around the shaft **412** along the longitudinal axis A_L. Each of the rotatable sleeves **422** cooperates with a different pair of opposing actuation elements **226'**, and the axial spacing of the engagement features **230** of the actuation elements is arranged to achieve respective cooperation with the rotatable sleeves **422**, as explained further below. Each of the rotatable sleeves **422** includes a flange **424** extending radially from an exterior of the rotatable sleeve **422**. Finally, the mechanical linkage of the force transmission mechanism **410** further comprises a connecting rod **420** coupling a respective flange **424** and respective crank arm **418**. Thus, each drive input shaft **416** is operably coupled to a respective pair of opposing actuation elements (such as actuation elements **226'**, not shown in FIG. **4**) of the shaft **412**.

[0047] In various embodiments, a rotatable sleeve **422** may be coupled to fewer or more than two actuation elements **226'**. For example, a single rotatable sleeve may be coupled to one, two, three or more actuation elements **226'**. Further, different rotatable sleeves may be coupled to a different number of actuation elements **226'**. For example, a first rotatable sleeve **416***a* may be coupled to two actuation elements **226'**, while a second rotatable sleeve **416***b* may be coupled to one actuation element **226'**. The numbers and arrangements depicted in the various illustrations herein are exemplary and non-limiting.

[0048] Each of the rotatable sleeves **422** can include features configured to interact with engagement features **230** (FIG. **3**) of the actuation elements **226'** (FIG. **3**). For example, each of the rotatable sleeves **422** can be configured to drive an opposing pair of actuation elements **226'** to articulate the shaft **412** in a degree of freedom with which the opposing pair of actuation elements **226'** is associated.

[0049] Reference is now made to FIG. **5**A, which shows a perspective view of a rotatable sleeve **422**, and FIG. **5**B, which shows a projection of the interior surface of the rotatable sleeve **422** onto the plane of FIG. **5**B (i.e., the interior surface of the rotatable sleeve **422** "unrolled" onto the plane of the drawing). Each of the rotatable sleeves **422** includes a first cam surface feature **532** and a second cam surface feature **534**. Each of the first and second cam surface features **532** and **534** can be configured as a recessed slot on an interior wall (e.g., bore) **536** of the rotatable sleeve **422** arranged and dimensioned to receive and engage an individual engagement feature **230** (FIG. **3**) of the actuation element **226'**. More specifically, the cam surface features **532** and **534** can be configured to respectively receive the engagement features **230** (FIG. **3**) of the pair of opposing actuation elements **226'** with which the sleeve **422** cooperates. The cam surface features **532** and **534** are in the form of inclined recesses angled, for example at least partially spiraling (following a helical path), relative to the longitudinal axis in opposite directions. That is, when viewed along the longitudinal axis AL of the shaft **212**, the cam surface features **532** and **534** rotate in opposite directions about the longitudinal axis AL of the shaft **212** as the sleeve **422** rotates.

[0050] The pitch of the first and second cam surface features **532** and **534** can be chosen based on the overall desired translational movement of the actuation elements **226'**, the range of motion of the drive input shafts **416**, the force required to move the actuation elements **226'**, and any requirement that the actuation elements **226'** be back-drivable (i.e., whether an external force applied to the shaft **212** can articulate the shaft **212**), and other factors. In some exemplary embodiments, the first and second cam surface features **532** and **534** follow a helical path. In alternate embodiments, a rotatable sleeve **422** may have fewer or more than two cam surface features. For example, a rotatable sleeve **422** may have a single cam surface feature to couple to a single actuation element **226'**, or a rotatable sleeve **422** may have three or more cam surface features to enable coupling to three or more respective actuation elements **226'**. The angle of the helical pitch of the cam surface features relative to the axis of the sleeve can be selected based on various instrument specifications, such as torque, space constraints, and other design criteria those having ordinary skill in the art would be familiar with. Differing pitches can affect the speed at which the actuation element coupled thereto moves and cam surface features within a sleeve can have the same or different pitches, and likewise with cam surface features of different sleeves.

[0051] FIGS. **6** and **7** show isolated views of the drive input shaft **416** and crank arm **418**. The drive input shaft **416** (FIG. **6**) includes a drive input disc **413** configured to engage a drive output of a manipulator system as discussed above. Alternatively, the drive input shaft **416** could be configured to be driven by a manual input. The shaft **416** includes the resilient, radially inwardly deflectable clip **423** at the free end portion of the shaft **416**. The shaft **416** has a cut out cross-section at the distal end portion thereof which is configured to mate with a complementary cross-section of a bore **426** of a hub portion **421** of the crank arm **418**. For example, the bore **426** and the shaft **416** can each have a flat surface portion that interface with each other to provide an anti-rotational interface (e.g., D-shaped cross sections) between the two components. Upon insertion of the shaft **416** within the bore **426** of the crank arm **418**, the surface of the bore **426** engages a protrusion **423'** on the resilient clip **423** which has an angled surface allowing the hub portion **421** of the crank arm **418** to slide up the distal end portion of the shaft **416**, deflecting the clip **423**. Once the hub portion **421** passes the protrusion **423**, the resilient clip **423** deflects back in place and the protrusion **423** provides a stop surface preventing the crank arm **418** from sliding axially off the distal end portion of the shaft **416** and thereby retaining the crank arm **418** on the shaft **419**, as shown in FIG. **4**.

[0052] The cross-sectional configurations of the bore **426** of the crank arm **418** and the distal portion of the shaft **416** provides an interface that orients the shaft **416** and crank arm **418** relative to each other and prevents relative rotation between the shaft **416** and the crank arm **418**. The configuration of the drive input shaft **416** and crank arm **418** described above may contribute to low part count and ease of assembly of the force transmission mechanism **410**. As noted above, the particular configurations of the bore and shaft cross-sections are not required and various other arrangement for coupling the drive input shaft and crank arm in a rotationally fixed relationship is within the scope of the disclosure, such as a splined interface, keyed interface, a pin interface such as roll pin, or other arrangements as would be familiar to those having ordinary skill in the art. Further, while the drive input shaft **416** and crank arm **418** are shown and described as separate components, in other embodiments, the drive input shaft **416** and crank arm **418** can optionally be formed integrally or in other configurations as would be apparent to those having ordinary skill in the art. Further, as described above, additional or alternative connections may be used in conjunction with or alternative to the resilient clip **423**, such as such as fasteners, adhesive, a press fit, or other connection.

[0053] In use, rotation of a drive input shaft **416** results in concurrent rotation of a rotatable sleeve **422** with which the drive input shaft **416** is associated via a respective connecting rod **420**. Rotation of the rotatable sleeve **422** causes the cam surface features **532**, **534** to bear against the respective engagement features **230** of the actuation elements **226'**, thereby causing translational movement of the pair of actuation elements **226'** with which the rotatable sleeve **422** cooperates. For example, rotation of the rotatable sleeve **422** causes a first one of the pair of actuation elements **226'** to translate in a first direction (e.g., proximally), and a second one of the pair of actuation elements **226'** to translate in a second direction opposite to the first direction (e.g., distally) based on the direction of the helical recesses and the rotational direction of the drive input shaft **416**, resulting in articulation of the shaft **212** in the degree of freedom with which the pair of actuation elements **226'** is associated (e.g., pitch or yaw). Articulation of the shaft **412** in the same degree of freedom, but in the opposite direction, can be achieved by reversing the direction of rotation of the drive input shaft **416** associated with that degree of freedom so that rotation of the rotatable sleeve **422** and translational movement of the corresponding opposing pair of actuation elements **226'** is reversed.

[0054] As will be appreciated by one of ordinary skill in the art, rotation of the rotatable sleeves **422** imparts loads on the engagement features **230** not only in the direction of desired movement of the actuation elements **226'**, i.e., along the longitudinal axis A_L of the shaft **212**, but also in lateral directions perpendicular to the longitudinal axis A_L of the shaft **212**. The force transmission mechanism **410** (FIG. **4**) can include additional structural features configured to support the actuation elements **226'** and lessen deflection of the actuation elements **226'** resulting from such side-loading. For example, referring now to FIG. **8**, the shaft **212** may be provided with a tubular support member **838**. The tubular support member **838** is configured to surround the actuation elements **226'** (FIG. **3**) in the region of the engagement features **230** and includes slots **840** elongated in the axial direction of the shaft **212** and through which the engagement features **230** pass. In the assembled state shown in FIG. **4**, the rotatable sleeves **422** surround the tubular support member **838**, and the engagement features **230** protrude through the slots **840** and into the cam surface features **532** and **534** of each rotatable sleeve **422**. The slots **840** constrain movement of the engagement features **230** to move longitudinally (parallel to the axial direction A_L) while supporting the engagement features **230** and actuation elements **226'** from side loads resulting from rotation of the rotatable sleeves **422**.

[0055] Due to the presence of the tubular support member **838** and the opposite helical directions of the cam surface features **532** and **534**, the engagement features **230** have constrained motion within the assembly. Various components of the force transmission mechanism **410** can include features configured to facilitate assembly of the components. For example, each of the rotatable sleeves **422** (FIG. **5**A) can include one or more assembly ports **542** through which the engagement features **230** may be pushed and pass into the longitudinal slots **840** of the tubular support member **838** and then into one of a series of holes provided in the actuation elements **226'**. The engagement features **230** can be separate from the actuation elements **226'** and pressed into one of the series of holes in the actuation elements with a press fit, an interference fit, or other means of connection to the actuation elements **226'**. In other embodiments, the engagement features **230** may be made integrally with the actuation elements. Rotation of the sleeves **422** can then occur until the engagement feature is positioned in a corresponding cam surface feature **532**, **534**.

[0056] In the embodiment of FIGS. **4** and **5**, the rotatable sleeves **422** are rotatable members configured with a common axis of rotation generally coaxial with the longitudinal axis A_L of the shaft **212**. In other embodiments, actuator assemblies of the force transmission mechanism can include rotatable members that do not share a common axis of rotation and do not rotate about axes parallel to or coaxial with the longitudinal axis of the shaft **212**.

[0057] For example, with reference now to FIG. **9**, another embodiment of a force transmission mechanism **910** according to the present disclosure is shown. In this embodiment, a rotatable coupler mechanism comprising a gimbal is utilized to transmit rotational movement of the rotary drive input to the translational motion of push-pull actuation elements. More specifically, a pair of actuation elements **926** that work together in tandem to create the opposite pushing/pulling forces can be coupled to differing locations of a rotatable gimbal **944**. Similar to the embodiment of FIGS. **4**-**8**, the embodiment of FIG. **9** can include eight actuation elements **926** to actuate four independent degrees of freedom of the shaft **912** via translational movement of each of a pair of actuation elements **926** working in a coordinated fashion in response to rotation of the gimbal. But, as discussed above, any number of actuation elements can be utilized to articulate a degree of freedom, including a single actuation element, to more than two for any particular degree of freedom.

[0058] In this embodiment, translation of a pair of actuation elements **926'** is accomplished by rotating the gimbal **944** about an axis intersecting the longitudinal axis of the shaft A_L and perpendicular to a line extending through the actuation elements **926'** desired to be actuated. For example, in the embodiment of FIG. **9**, if labeled actuation elements **926'** are desired to be actuated, the gimbal **944** is rotated about axis A_G, thereby causing longitudinal movement of the actuation elements **926** in opposite directions. In the embodiment of FIG. **9**, the gimbal **944** can be rotatable about two perpendicular axes defined by the locations at which the actuation elements **926'** are attached to the respective gimbal **944**. Similar to the arrangement of the rotatable sleeves in the above embodiments, plural gimbals **944** could be used and at differing locations long the longitudinal axis of the shaft to engage with and transmit force to different ones or sets of actuation elements. Depending on the location of the actuation elements about the longitudinal axis A_L, articulation of an articulable segment can occur in differing directions and about differing axes (degrees of freedom).

[0059] The gimbal **944** is rotated in the desired manner via yoke **946**, with. each gimbal **944** being rotated by two rotatable yokes **946** (only one yoke **946** shown in FIG. **9**) to actuate the two degrees of freedom associated with each gimbal **944**. Each rotatable yoke **946** is coupled to a driveshaft **948**, which is in turn operatively coupled to a drive input **916** by a bevel gear set **955**. Rotation of the drive input **916** drives rotation of the driveshaft **948** and yoke **946**, thereby generating longitudinal movement of the actuation elements **926** in opposite directions to articulate the shaft **212** (FIG. **2**). In some embodiments, multiple driveshafts **948** can be provided in a nested coaxial configuration depending on the number of degrees of freedom of the shaft to be articulated. As with other embodiments described above, the shaft **912** of FIG. **9** can include an inner layer **907** configured to impart roll motion to the shaft **912**. More specifically, the inner layer **907** can be coupled to a drive input **916** through a series of gears **950** (e.g., planetary and stacked gear train in the embodiment of FIG. **9**) and rotary input can be transmitted from the roll drive input **916** through the gear train **950** to impart roll motion to the inner layer **907** and thereby the shaft **912**.

[0060] Force transmission systems according to embodiments of the present disclosure provide compact and relatively simple systems that can be adapted to various existing manipulator and shaft architecture. Features of force transmission systems according to the present disclosure can facilitate manufacturing, assembly, and contribute to generally low part count and relatively high reliability. Further, embodiments of force transmission mechanisms discussed herein can be used with actuation elements other than actuation elements **226'**, **926'** discussed herein. For example, embodiments of force transmission mechanisms as discussed herein can optionally be used with cable-type actuation elements, such as pull-pull type actuation elements, compression-type actuation elements (e.g., rod elements), or other types of actuation elements.

[0061] Embodiments described herein may be used, for example, with remotely operated, computer-assisted systems (such, for example, teleoperated surgical systems) such as those described in, for example, U.S. Pat. No. 9,358,074 (filed May 31, 2013) to Schena et al., entitled "Multi-Port Surgical Robotic System Architecture"; and U.S. Pat. No. 9,295,524 (filed May 31, 2013) to Schena et al., entitled "Redundant Axis and Degree of Freedom for Hardware-Constrained Remote Center Robotic Manipulator", each of which is hereby incorporated by reference in its entirety. Further, embodiments described herein may be used, for example, with any of the da Vinci® Surgical System commercialized by Intuitive Surgical, Inc., of Sunnyvale, California.

[0062] The embodiments described herein are not limited to the surgical systems noted above, and various other teleoperated, computer-assisted surgical system configurations may be used with the embodiments described herein. Further, although various embodiments described herein are discussed in connection with a manipulating system of a teleoperated surgical system, the present disclosure is not limited to use with a teleoperated surgical system. Various embodiments described herein can optionally be used in conjunction with hand-held, manual instruments.

[0063] As discussed above, in accordance with various embodiments, transmission systems of the present disclosure are configured for use in teleoperated, computer-assisted surgical systems employing robotic technology (sometimes referred to as robotic surgical systems). Referring now to FIG. **10**, an embodiment of a manipulator system **1000** of a computer-assisted surgical system, to which surgical instruments are configured to be mounted for use, is shown. Such a surgical system may further include a user control system, such as a surgeon console (not shown) for receiving input from a user to control instruments coupled to the manipulator system **1000**, as well as an auxiliary system, such as auxiliary systems associated with the da Vinci® surgical systems noted above.

[0064] As shown in the embodiment of FIG. **10**, a manipulator system **1000** includes a base **1020**, a main column **1040**, and a main boom **1060** connected to main column **1040**. Manipulator system **1000** also includes a plurality of manipulator arms **1010**, **1011**, **1012**, **1013**, which are each connected to main boom **1060**. Manipulator arms **1010**, **1011**, **1012**, **1013** each include an instrument mount portion **1022** to which an instrument **1030** may be mounted, which is illustrated as being attached to manipulator arm **1010**. While the manipulator system **1000** of FIG. **10** is shown and described having a main boom **1060** to which the plurality of manipulator arms are coupled and supported thereby, in other embodiments, the plurality of manipulator arms can be coupled and supported by other structures, such as an operating table, a ceiling, wall, or floor of an operating room, etc.

[0065] Instrument mount portion **1022** comprises a drive assembly **1023** and a cannula mount **1024**, with a transmission mechanism **1034** (which may generally correspond to the force transmission mechanisms **110**, **410**, **910** discussed in connection with FIGS. **1**, **4**, and **9**) of the instrument **1030** connected to the drive assembly **1023**, according to an embodiment. Cannula mount **1024** is configured to hold a cannula **1036** through which a shaft **1032** of instrument **1030** may extend to a surgery site during a surgical procedure. Drive assembly **1023** contains a variety of drive and other mechanisms that are controlled to respond to input commands at the surgeon console and transmit forces to the transmission mechanism **1034** to actuate the instrument **1030**. Although the embodiment of FIG. **10** shows an instrument **1030** attached to only manipulator arm **1010** for ease of viewing, an instrument may be attached to any and each of manipulator arms **1010**, **1011**, **1012**, **1013**.

[0066] Other configurations of surgical systems, such as surgical systems configured for single-port surgery, are also contemplated. For example, with reference now to FIG. **11**, a portion of an embodiment of a manipulator arm **2140** of a manipulator system with two surgical instruments **2300**, **2310** in an installed position is shown. The surgical instruments **2300**, **2310** can generally correspond to instruments discussed above, such as instrument **100** disclosed in connection with FIG. **1**. The schematic illustration of FIG. **11** depicts only two surgical instruments for simplicity, but more than two surgical instruments may be mounted in an installed position at a manipulator system as those having ordinary skill in the art are familiar with. Each surgical instrument **2300**, **2310** includes a shaft **2320**, **2330** that at a distal end has a moveable end effector or an endoscope, camera, or other sensing device, and may or may not include a wrist mechanism (not shown) to control the movement of the distal end.

[0067] In the embodiment of FIG. **11**, the distal end portions of the surgical instruments **2300**, **2310** are received through a single port structure **2380** to be introduced into the patient. As shown, the port structure includes a cannula and an instrument entry guide inserted into the cannula. Individual instruments are inserted into the entry guide to reach a surgical site.

[0068] Other configurations of manipulator systems that can be used in conjunction with the present disclosure can use several individual manipulator arms. In addition, individual manipulator arms may include a single instrument or a plurality of instruments. Further, as discussed above, an instrument may be a surgical instrument with an end effector or may be a camera instrument or other sensing instrument utilized during a surgical procedure to provide information, (e.g., visualization, electrophysiological activity, pressure, fluid flow, and/or other sensed data) of a remote surgical site.

[0069] Force transmission mechanisms **2385**, **2390** (which may generally correspond to force transmission mechanism **110** disclosed in connection with FIG. **1**, force transmission mechanism **410** disclosed in connection with FIG. **4**, and force transmission mechanism **910** disclosed in connection with FIG. **9**) are disposed at a proximal end of each shaft **2320**, **2330** and connect through a sterile adaptor **2400**, **2410** with drive assemblies **2420**, **2430**. Drive assemblies **2420**, **2430** contain a variety of internal mechanisms (not shown) that are controlled by a controller (e.g., at a control cart of a surgical system) to respond to input commands at a surgeon side console of a surgical system to transmit forces to the force transmission mechanisms **2385**, **2390** to actuate surgical instruments **2309**, **2310**.

[0070] The embodiments described herein are not limited to the embodiments of FIG. **10** and FIG. **11**, and various other teleoperated, computer-assisted surgical system configurations may be used with the embodiments described herein. The diameter or diameters of an instrument shaft and end effector are generally selected according to the size of the cannula with which the instrument will be used and depending on the surgical procedures being performed.

[0071] This description and the accompanying drawings that illustrate various embodiments should not be taken as limiting. Various mechanical, compositional, structural, electrical, and operational changes may be made without departing from the scope of this description and the invention as claimed, including equivalents. In some instances, well-known structures and techniques have not been shown or described in detail so as not to obscure the disclosure. Like numbers in two or more figures represent the same or similar elements. Furthermore, elements and their associated features that are described in detail with reference to one embodiment may, whenever practical, be included in other embodiments in which they are not specifically shown or described. For example, if an element is described in detail with reference to one embodiment and is not described with reference to another embodiment, the element may nevertheless be claimed as included in the other embodiment.

[0072] For the purposes of this specification and appended claims, unless otherwise indicated, all numbers expressing quantities, percentages, or proportions, and other numerical values used in the specification and claims, are to be understood as being modified in all instances by the term "about," to the extent they are not already so modified. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

[0073] It is noted that, as used in this specification and the appended claims, the singular forms "a," "an," and "the," and any singular use of any word, include plural referents unless expressly and unequivocally limited to one referent. As used herein, the term "include" and its grammatical variants are intended to be non-limiting, such that recitation of items in a list is not to the exclusion of other like items that can be substituted or added to the listed items.

[0074] Further, this description's terminology is not intended to limit the invention. For example, spatially relative terms—such as "beneath", "below", "lower", "above", "upper", "proximal", "distal", and the like—may be used to describe one element's or feature's relationship to another element or feature as illustrated in the figures. These spatially relative terms are intended to encompass different positions (i.e., locations) and orientations (i.e., rotational placements) of a device in use or operation in addition to the position and orientation shown in the figures. For example, if a device in the figures is turned over, elements described as "below" or "beneath" other elements or features would then be "above" or "over" the other elements or features. Thus, the exemplary term "below" can encompass both positions and orientations of above and below. A device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0075] Further modifications and alternative embodiments will be apparent to those of ordinary skill in the art in view of the disclosure herein. For example, the devices and methods may include additional components or steps that were omitted from the diagrams and description for clarity of operation. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the present teachings. It is to be understood that the various embodiments shown and described herein are to be taken as exemplary. Elements and materials, and arrangements of those elements and materials, may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features of the present teachings may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of the description herein. Changes may be made in the elements described herein without departing from the spirit and scope of the present teachings and following claims.

[0076] It is to be understood that the particular examples and embodiments set forth herein are non-limiting, and modifications to structure, dimensions, materials, and methodologies may be made without departing from the scope of the present teachings.

[0077] Other embodiments in accordance with the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with the following claims being entitled to their fullest breadth, including equivalents, under the applicable law.

What is claimed is:

1. A steerable instrument, comprising:

a shaft extending along a longitudinal axis from a proximal end portion to a distal end portion, the shaft comprising an articulable segment;

a force transmission mechanism at the proximal end portion of the shaft, the force transmission mechanism comprising one or more rotatable drive components having axes of rotation spaced from the longitudinal axis of the shaft, the one or more rotatable drive components configured to be rotatably driven by respective drive input torques;

one or more push-pull actuation elements coupled to the articulable segment and configured to translate to transmit compressive forces to the articulable segment to articulate the articulable segment; and

a rotatable coupler mechanism coupling one of the one or more rotatable drive components to the one or more push-pull actuation elements, the rotatable coupler mechanism configured to rotate about the longitudinal axis of the shaft in response to rotation of the one of the one or more rotatable drive components to cause translation of the one or more push-pull actuation elements.

2. The steerable instrument of claim **1**, wherein the rotatable coupler mechanism comprises a cam surface that drives the one or more push-pull actuation elements to translate the one or more push-pull actuation elements.

3. The steerable instrument of claim **2**, wherein rotatable coupler mechanism comprises an inclined recess on an interior wall of the rotatable coupler mechanism.

4. The steerable instrument of claim **2**, further comprising:

a plurality of push-pull actuation elements; and

a plurality of cam surfaces configured to drive the plurality of push-pull actuation elements,

wherein the plurality of cam surfaces are configured to drive at least some of the plurality of push-pull actuation elements at differing rates from each other.

5. The steerable instrument of claim **1**, wherein the rotatable coupler mechanism comprises:

a first rotatable sleeve comprising a first plurality of cam surfaces engageable with a first pair of push-pull actuation elements and configured to drive translation of the first pair push-pull actuation elements in opposite directions in response to rotation of the first rotatable sleeve, and

a second rotatable sleeve comprising a second plurality of cam surfaces engageable with a second pair of push-pull actuation elements to drive translation of the second pair of push-pull actuation elements in opposite directions in response to rotation of the second rotatable sleeve, the second rotatable sleeve coupled to another of the one or more rotatable drive components.

6. The steerable instrument of claim **5**, wherein the first rotatable sleeve and the second rotatable sleeve are each coaxial with the longitudinal axis of the shaft, the first rotatable sleeve being distal to the second rotatable sleeve.

7. The steerable instrument of claim **5**,

wherein the first and second plurality of cam surfaces comprise respective first and second inclined recesses on an interior wall of the first rotatable sleeve and the second rotatable sleeve, respectively,

wherein each of the first pair of push-pull actuation elements comprises a protrusion configured to be received within the first inclined recess, and

wherein each of the second pair of push-pull actuation elements comprises a protrusion configured to be received within the second inclined recess.

8. The steerable instrument of claim **1**, wherein the rotatable coupler mechanism comprises a gimbal which is coupled to a plurality of the push-pull actuation elements.

9. The steerable instrument of claim **8**, wherein:

the one or more push-pull actuation elements comprises a first pair of push-pull actuation elements;

the steerable instrument comprises a second pair of push-pull actuation elements;

the first pair of push-pull actuation elements is coupled to the gimbal at a first pair of locations on the gimbal such that rotation of the gimbal about a first axis actuates the first pair of push-pull actuation elements; and

the second pair of push-pull actuation elements is coupled to the gimbal at a second pair of locations on the gimbal such that rotation of the gimbal about another axis actuates the second pair of push-pull actuation elements.

10. A steerable instrument, comprising:

a shaft extending along a longitudinal axis from a proximal end portion to a distal end portion, the shaft comprising an articulable segment;

a first actuation element operably coupled to the articulable segment of the shaft and extending along the shaft from the articulable segment to the proximal end portion of the shaft;

a second actuation element operably coupled to the articulable segment of the shaft and extending along the shaft from the articulable segment to the proximal end portion of the shaft; and

a rotatable coupler mechanism rotatable about the longitudinal axis of the shaft and comprising cam surfaces respectively engageable with the first and second actuation elements so as to drive the first and second actuation elements in translation in response to rotation of the rotatable coupler mechanism.

11. The steerable instrument of claim **10**, wherein the articulable segment is articulatable in response to translation of the first actuation element and the second actuation element in opposite directions to each other.

12. The steerable instrument of claim **10**, further comprising a third actuation element and a fourth actuation element, the third actuation element and the fourth actuation element coupled to the articulable segment of the shaft.

13. The steerable instrument of claim **12**, wherein:

the rotatable coupler mechanism comprises a first rotatable sleeve comprising first cam surfaces engageable with the first and second actuation elements and a second rotatable sleeve comprising second cam surfaces engageable with the third and fourth actuation elements.

14. The steerable instrument of claim **13**, wherein:

the articulable segment is articulable in a first degree of freedom based on translation of the first actuation element and the second actuation element in opposite directions to each other; and

the articulable segment is articulable in a second degree of freedom, differing from the first degree of freedom, based on translation of the third actuation element and the fourth actuation element in opposite directions to each other.

15. The steerable instrument of claim **10**, further comprising a rotatable drive component having an axis of rotation offset from the longitudinal axis of the shaft, wherein the rotatable coupler mechanism is operably coupled to be driven in rotation in response to rotation of the rotatable drive component.

16. The steerable instrument of claim **15**, wherein the rotatable coupler mechanism comprises a rotatable sleeve comprising the cam surfaces and operably coupled to the rotatable drive component.

17. The steerable instrument of claim **15**, wherein the rotatable drive component is a first rotatable drive component and the axis of rotation is a first axis of rotation, and wherein the steerable instrument further comprises a second rotatable drive component having a second axis of rotation offset from the first axis of rotation and the longitudinal axis.

18. The steerable instrument of claim **17**, wherein the rotatable coupler mechanism comprises:

a first rotatable sleeve operably coupled to be driven in rotation by the first rotatable drive component and comprising a cam surface engageable with the first actuation element, and

a second rotatable sleeve operably coupled to be driven in rotation by the second rotatable drive component and comprising another cam surface engageable with the second actuation element.

19. The steerable instrument of claim **18**, wherein:

a first cam surface of the rotatable coupler mechanism is operably coupled to the first actuation element and is configured to be driven in rotation by the first rotatable drive component, and

a second cam surface of the rotatable coupler mechanism is operably coupled to second actuation element and is configured to be driven in rotation by the second rotatable drive component.

20. The steerable instrument of claim **15**, wherein the cam surfaces comprise a first recess and a second recess engageable with respective first and second protrusions of the first and second actuation elements.

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